Tutorial 2

Learning, cascades, and games

with step-by-step solutions

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How this tutorial works

- Four blocks. Each one opens with a compact recap of the theory and a pointer to the lecture where it is developed, and then moves to exercises on that same material.
- The exercises use the same templates as the online quizzes on Virtuale and as the mid-term. Same mechanics, different numbers.
- Try each one first. We solve it together afterwards.
- Questions on the *content* of the course go to Prof. Bigoni. Questions on the organisation, the quizzes and the exam come to me: `marco.rosso4@unibo.it`

Coverage: Lectures 5, 6, 7, 8.

Plan

1. Learning and information cascades
2. Simultaneous games: prisoner's dilemma and coordination
3. Expectations, guessing games, mixed strategies
4. Multi-stage games

Sample *oral* questions at the end.

PART 1

Learning and information cascades

Belief learning and reinforcement learning

Lecture 5 – "Belief learning", "Reinforcement learning", "Belief vs. Reinforcement learning"

$$\text{belief learning} \qquad \Pr(L) = \frac{\alpha + N_L}{\alpha + \alpha + N} \qquad N = N_L + N_R$$

$$\text{reinforcement learning} \qquad \Pr(L) = \frac{\alpha + e_L}{2\alpha + e_L + e_R}$$

- α determines how quickly probabilities respond: a large α keeps them close to $1/2$.
- Belief learning: N_L, N_R are the **observations** of each event. The left-hand side is a belief.
- Reinforcement learning: e_L, e_R are the total **earnings** from predicting each event. The left-hand side is a choice probability.

Exercise 1.1

Learning

(a) In a binary prediction task, a subject observes the sequence of events

$$L,\ L,\ L,\ L,\ R.$$

Using the belief-learning model with $\alpha = 2$, compute $\Pr(L)$ after these five observations.

(b) Another subject earns €1 for each correct prediction, nothing otherwise. After 10 trials she has predicted L 7 times (5 of them correct) and R 3 times (1 correct). Under reinforcement learning with $\alpha = 2$, what is the probability that she predicts L in trial 11?

Exercise 1.1 – step by step

(a) Belief learning

Step 1. Count the observations: $L, L, L, L, R \Rightarrow N_L = 4, N_R = 1, N = 5$.

Step 2. Write the formula: $\Pr(L) = \dfrac{\alpha + N_L}{\alpha + \alpha + N}$.

Step 3. Substitute $\alpha = 2$: numerator $2 + 4 = 6$, denominator $2 + 2 + 5 = 9$.

Step 4. $\Pr(L) = 6/9 = 2/3 \approx 0.667$. Check: $\Pr(R) = (2 + 1)/9 = 3/9$, and $6/9 + 3/9 = 1$.

(b) Reinforcement learning

Step 1. Compute *earnings*, not counts. L predicted 7 times, 5 correct $\times$ €1 $\Rightarrow e_L = 5$. R predicted 3 times, 1 correct $\Rightarrow e_R = 1$. Wrong predictions earn 0.

Step 2. $\Pr(L) = \dfrac{\alpha + e_L}{2\alpha + e_L + e_R} = \dfrac{2 + 5}{4 + 5 + 1} = \dfrac{7}{10} = 0.70$. Check: $\Pr(R) = 3/10$.

Typical mistake: using $e_L = 7$ (times L was predicted) instead of $e_L = 5$ (earnings from L).

Information cascades

Lecture 5 – "The basic situation" to "Incorrect cascades"

Two cups, equally likely. Each person sees **one private draw** and **all previous decisions**, not previous draws. Paid only for guessing the cup.

- 1st person: follows own draw.
  $\Pr(A \mid a) = 2/3$
- 2nd person: in a conflict, the two draws cancel out $\Rightarrow$ goes with own draw
- From the 3rd on: two matching decisions outweigh one's own draw $\Rightarrow$ **cascade**

prior	own	$\Pr(A)$	decision
A, A	*a*	0.89	A
A, A	*b*	0.67	A *(cascade)*
A, B	*a*	0.67	A
A, B	*b*	0.33	B

Incorrect cascade from the first two draws: $\left(\frac{1}{3}\right)\left(\frac{1}{3}\right) = \frac{1}{9}$. In a cascade, decisions convey **no information** about private draws $\Rightarrow$ cascades may be fragile.

Exercise 1.2

Information cascades

Cup A contains **3 Amber and 1 Blue**; Cup B contains **1 Amber and 3 Blue**. Each cup is equally likely. People decide in sequence, each seeing one private draw (with replacement) and all previous decisions.

(a) The first person draws amber. What is her probability of Cup A?
(b) Complete the table for the third person, assuming that the first two followed their own draws:

prior decisions	own draw	Pr(Cup A)	decision
A, A	*a*	_____	____
A, A	*b*	_____	____
A, B	*a*	_____	____
A, B	*b*	_____	____

(c) Which row starts a cascade?
(d) What is the probability that the first two people both receive a misleading draw, and so start an incorrect cascade?

Exercise 1.2 – step by step (a, b)

(a) **Step 1.** There are 4 amber marbles, 3 in Cup A and 1 in Cup B. With equal priors every marble is equally likely to be drawn, so $\Pr(A \mid a) = 3/4$. Same result with Bayes: $\dfrac{(3/4)(1/2)}{(3/4)(1/2) + (1/4)(1/2)} = \dfrac{3}{4}$.

(b) **Step 1.** The first two followed their draws, so decision A reveals a and decision B reveals b.
Step 2. Likelihood of each draw: $\Pr(a \mid A) = \frac{3}{4}$, $\Pr(b \mid A) = \frac{1}{4}$, $\Pr(a \mid B) = \frac{1}{4}$, $\Pr(b \mid B) = \frac{3}{4}$.
Step 3. Priors are equal and cancel: $\Pr(A \mid S) = \Pr(S \mid A)/[\Pr(S \mid A) + \Pr(S \mid B)]$.

prior	own	$\Pr(S \mid A)$	$\Pr(S \mid B)$	$\Pr(A \mid S)$
A, A	a	$(3/4)^3 = 27/64$	$(1/4)^3 = 1/64$	$27/28 = 0.96$
A, A	b	$(3/4)^2(1/4) = 9/64$	$(1/4)^2(3/4) = 3/64$	$9/12 = 0.75$
A, B	a	$(3/4)(1/4)(3/4) = 9/64$	$(1/4)(3/4)(1/4) = 3/64$	$9/12 = 0.75$
A, B	b	$(3/4)(1/4)^2 = 3/64$	$(1/4)(3/4)^2 = 9/64$	$3/12 = 0.25$

Step 4. Decide for the cup with posterior above $1/2$: A, A, A, B.

Exercise 1.2 – step by step (c, d)

(c) **Step 1.** Look for the row where the decision goes *against* the person's own draw: row 2 (own draw ***b***, decision A).

Step 2. From then on, the 4th person sees A, A, A: the third A carries no information (it was taken regardless of the draw), so she is in the same situation as the 3rd person and also chooses A. The cascade continues.

Intuition, as in the lecture: one ***a*** and one ***b*** cancel out, so only the *net* number of revealed draws matters. In row 2 the net count is one ***a***, hence the same **3/4** as a single amber draw.

(d) **Step 1.** An incorrect cascade starts if both of the first two people draw the colour that points to the wrong cup.

Step 2. Whichever cup is used, it contains 1 marble of the "wrong" colour out of 4: probability **1/4** per draw.

Step 3. Draws are independent (with replacement): $\frac{1}{4} \times \frac{1}{4} = \frac{1}{16} = \mathbf{0.0625}$.

Step 4. Compare with the lecture's cups (1 out of 3): $\frac{1}{3} \times \frac{1}{3} = \frac{1}{9} \approx \mathbf{0.111}$. More informative draws, fewer incorrect cascades.

PART 2

Simultaneous games

Nash equilibrium and the prisoner's dilemma

Lecture 6 – "A game", "Nash equilibrium", "The prisoner's dilemma", "Backward induction"

Nash equilibrium: a set of strategies, one for each player, such that nobody would wish to deviate, given the strategies used by the others.

Method: underline each player's best response to each strategy of the other. A cell with both payoffs underlined is a NE.

	Y	Z
Y	$\underline{3}, \underline{3}$	$\underline{10}, 0$
Z	$0, \underline{10}$	$8, 8$

- Whatever the other does, Y is better: (Y,Y) is the only NE, although (Z,Z) is better for both
- Finite, known number of rounds: backward induction $\Rightarrow$ no cooperation, even in the first round

Lab: random matching, 43% cooperation early $\rightarrow$ about 20% in rounds 15–20. Partner matching: more cooperation, which drops as the final round approaches (*end-game effect*).

Exercise 2.1

A game

Consider a game with the following payoff matrix.

	Left	Right
Top	35, 35	10, 45
Bottom	45, 10	20, 20

(a) Identify the pure-strategy equilibrium outcomes of this game: a. (35, 35) b. (10, 45) c. (45, 10) d. (20, 20)

(b) What type of game is this?

a. prisoners' dilemma b. battle of the sexes c. ultimatum game d. matching pennies game e. traveler's dilemma

Exercise 2.1 – step by step

(a) Step 1. Row's best responses, one column at a time:

- Column plays Left: Top gives 35, Bottom gives 45 $\Rightarrow$ Bottom; underline 45.
- Column plays Right: Top gives 10, Bottom gives 20 $\Rightarrow$ Bottom; underline 20.

Step 2. Column's best responses, one row at a time:

- Row plays Top: Left gives 35, Right gives 45 $\Rightarrow$ Right; underline 45.
- Row plays Bottom: Left gives 10, Right gives 20 $\Rightarrow$ Right; underline 20.

Step 3. The only cell with both payoffs underlined is (Bottom, Right) $= (20, 20)$: answer **d**.

(b) Step 1. Bottom is Row's best response to *both* columns, and Right is Column's best response to *both* rows.

Step 2. The equilibrium payoffs $(20, 20)$ are lower for both than $(35, 35)$.

Step 3. These two features define the **prisoners' dilemma** (a). The other options do not fit: a battle of the sexes has two pure equilibria, matching pennies has none, the ultimatum game is sequential.

Exercise 2.2

Repetition

Consider the game of Exercise 2.1.

(a) What is the Nash equilibrium of the game, when it is repeated for 10 periods with fixed pairs? Briefly illustrate the reasoning behind your answer.

(b) Consider an experiment in which the game is played for 10 periods, with fixed pairs. What is the behaviour that normally emerges in this type of situation, in the laboratory?

Exercise 2.2 – step by step

(a) Step 1. Start from the last period, 10. Nothing comes after it, so it is the one-shot game: both choose their dominant strategy, (Bottom, Right).

Step 2. Period 9. Period 10 will be (Bottom, Right) whatever happens in period 9, so cooperating in period 9 cannot be rewarded later: period 9 is also a one-shot game $\Rightarrow$ (Bottom, Right).

Step 3. Repeat the argument for periods 8, 7, …, 1.

Step 4. Result: (Bottom, Right) in every period, $10 \times 20 = 200$ each, against $10 \times 35 = 350$ with cooperation throughout.

Key condition: the number of periods is *finite and known*.

(b) Step 1. Early periods: a fair amount of cooperation (Top, Left).

Step 2. Last periods: cooperation drops (end-game effect).

Step 3. Fixed pairs (partner matching) sustain more cooperation than random matching, where it falls from about 43% to about 20%.

From production to coordination

Lecture 6 – “Alternative framing: co-workers”, “Coordination problems”, “Equilibrium selection”

Two workers, effort 0 or 1. The benefit per person depends on total effort; each unit of effort costs the worker who supplies it.

$$\text{payoff} = \text{benefit per person}(\text{total effort}) - \text{cost} \times \text{own effort}$$

total effort	0	1	2	cost	game
no synergy	3	10	18	10	prisoner’s dilemma: (Y,Y)
synergy	3	10	30	10	coordination: (Y,Y) and (Z,Z)
high cost	3	10	30	19	coordination, (Z,Z) very risky

Coordinating on the best equilibrium is not guaranteed: with a high cost of effort, behaviour is likely to converge to the equilibrium that is worst for all.

Exercise 2.3

Co-workers

Two co-workers each choose effort L (0) or H (1). The benefit per person depends on total effort:

total effort	0	1	2
benefit per person	€4	€12	€36

Each unit of effort costs €c to the worker who supplies it.

(a) Write the payoff matrix for $c = 12$ and find all pure-strategy Nash equilibria.

(b) Same, for $c = 20$.

(c) Now suppose there is no synergy: the benefit with total effort 2 is €20, and $c = 12$. What type of game is it?

Exercise 2.3 – step by step (a, b)

Step 1. Payoff = benefit per person (total effort) $- c \times$ own effort.

Step 2. Fill the matrix with $c = 12$, cell by cell:

- (L, L): total effort 0, benefit 4; nobody pays $\Rightarrow (4, 4)$
- (L, H): total effort 1, benefit 12; Row pays 0, Column pays 12 $\Rightarrow (12,\ 12 - 12) = (12, 0)$
- (H, L): symmetric $\Rightarrow (0, 12)$
- (H, H): total effort 2, benefit 36; each pays 12 $\Rightarrow (24, 24)$

Step 3. Best responses: if the other plays L, compare L 4 with H 0 $\Rightarrow$ L. If the other plays H, compare L 12 with H 24 $\Rightarrow$ H.

Step 4. Two NE: (L, L) and (H, H).

(b) Same steps with $c = 20$: (L, L) $= (4, 4)$; (L, H) $= (12,\ 12 - 20) = (12, -8)$; (H, H) $= (36 - 20,\ 36 - 20) = (16, 16)$. Other plays L: 4 against $-8 \Rightarrow$ L. Other plays H: 12 against 16 $\Rightarrow$ H. Again two NE, (L, L) and (H, H).

Exercise 2.3 – step by step (c)

Step 1. Only the (H, H) cell changes: benefit 20, each pays 12 $\Rightarrow$ $(8, 8)$. The other cells are as in (a): $(4, 4)$, $(12, 0)$, $(0, 12)$.

Step 2. Best responses: if the other plays L, compare L 4 with H 0 $\Rightarrow$ L. If the other plays H, compare L 12 with H 8 $\Rightarrow$ L.

Step 3. L is the best response whatever the other does: unique NE (L, L) $=$ $(4, 4)$, although (H, H) $=$ $(8, 8)$ is better for both $\Rightarrow$ **prisoner's dilemma**.

The comparison between (a) and (c) is the point of the lecture's co-worker example: the only difference is whether the second unit of effort makes the first one more productive (36 against 20). With synergy the game becomes a coordination game; without it, it is a prisoner's dilemma.

PART 3

Expectations, guessing games, mixed strategies

Expected payoff difference and basins of attraction

Lecture 7 – “Expectations and behavior”, “Learning and basin of attraction”, “Expected payoff difference”

p = probability that the other player chooses High.

$$\Delta = \mathbb{E}[\pi_H] - \mathbb{E}[\pi_L] \qquad \text{threshold: } \Delta = 0$$

game	Δ	threshold
(3,3), (10,0), (0,10), (20,20)	$13p - 3$	$p = 3/13 \approx 0.23$
(3,3), (10,−7), (−7,10), (13,13)	$13p - 10$	$p = 10/13 \approx 0.77$

- A risk-neutral player prefers High when p is above the threshold.
- **Basin of attraction** of an equilibrium: the beliefs for which learning pushes towards it. A higher cost of effort raises the threshold and shrinks the basin of the high-effort equilibrium.

Exercise 3.1

Thresholds

Take the two coordination games of Exercise 2.3, with $c = 12$ and $c = 20$. Let p be the probability that the other worker chooses H.

(a) For each game, write $\mathbb{E}[\pi_H]$, $\mathbb{E}[\pi_L]$ and Δ.

(b) How large should p be, for a risk-neutral worker to prefer H?

(c) What happens to the basin of attraction of (H,H) when the cost rises?

Exercise 3.1 – step by step

Step 1. Use Row's payoffs from Exercise 2.3. The other plays H with probability p, L with probability $1-p$.

Step 2. $c = 12$:

$$\mathbb{E}[\pi_H] = 24\,p + 0\,(1-p) = 24p \qquad \mathbb{E}[\pi_L] = 12\,p + 4\,(1-p) = 4 + 8p$$

$$\Delta = 24p - (4 + 8p) = 16p - 4$$

Step 3. $c = 20$:

$$\mathbb{E}[\pi_H] = 16\,p + (-8)(1-p) = 24p - 8 \qquad \mathbb{E}[\pi_L] = 4 + 8p \qquad \Delta = 16p - 12$$

Step 4. Set $\Delta = 0$: $16p = 4 \Rightarrow p = 1/4$; $16p = 12 \Rightarrow p = 3/4$.

Step 5. Check: $c = 12, p = \frac{1}{4}$: $\mathbb{E}[\pi_H] = 6, \mathbb{E}[\pi_L] = 4 + 2 = 6$. $c = 20, p = \frac{3}{4}$: $\mathbb{E}[\pi_H] = 18 - 8 = 10$, $\mathbb{E}[\pi_L] = 4 + 6 = 10$.

(c) Step 6. H is preferred for $p > 1/4$ with $c = 12$, but only for $p > 3/4$ with $c = 20$: the basin of attraction of (H, H) shrinks.

Guessing games

Lecture 7 – “The game”, “The Nash equilibrium”, “Experimental results”, “Introspection”

N players pick a number from 0 to 100. The winner is the one closest to $1/2$ of the average.

- The best response to any common decision above 0 is to go below it $\Rightarrow$ **the only Nash equilibrium is 0**.
- **Level-k**: level 0 does not think (randomises); level 1 best responds to level 0; level k best responds to level $k-1$. Most people are level 1 to 3.
- Lecture: three levels of iterated reasoning give $50 \to 25 \to 12.5 \to 6.25$.
- With repetition, people learn and choices converge towards the Nash equilibrium.

Exercise 3.2

Beauty contest

Five players each choose a number between 0 and 100. The winner is the player whose number is closest to **2/3** of the average of all numbers.

(a) What is the Nash equilibrium?

(b) Level-0 players choose at random, so that their average is 50. What do level-1, level-2 and level-3 players choose?

(c) The five choices are 40, 30, 25, 18 and 12. Who wins?

Exercise 3.2 – step by step

(a) **Step 1.** Suppose everybody chooses the same $x > 0$. The average is x, the target is $\frac{2}{3}x < x$: each player would win by going below the others. So no common $x > 0$ is an equilibrium.

Step 2. If everybody chooses 0, the target is 0 and all share the prize; anyone who deviates upwards moves away from the target. **NE: everybody chooses 0.**

(b) **Step 1.** Level 0: average 50.

Step 2. Level 1 best responds to 50: $\frac{2}{3} \times 50 = 33.3$.

Step 3. Level 2 best responds to 33.3: $\frac{2}{3} \times 33.3 = 22.2$.

Step 4. Level 3 best responds to 22.2: $\frac{2}{3} \times 22.2 = 14.8$.

(c) **Step 1.** Sum: $40 + 30 + 25 + 18 + 12 = 125$; average $= 125/5 = 25$.

Step 2. Target: $\frac{2}{3} \times 25 = 16.7$.

Step 3. Distances: $|40 - 16.7| = 23.3$, $|30 - 16.7| = 13.3$, $|25 - 16.7| = 8.3$, $|18 - 16.7| = 1.3$, $|12 - 16.7| = 4.7 \Rightarrow$ **18 wins**.

Exercise 3.3

A two-player game

The Row player can choose between “up” and “down”; the Column player can choose between “left” and “right”.

- If Row chooses up and Column chooses left: Row earns 8, Column earns 2.
- If Row chooses down and Column chooses right: Row earns 4, Column earns 6.
- In all other cases both players earn 0.

(a) Write the payoff table for this game.

(b) Select which category this game belongs to: a. prisoner’s dilemma b. matching pennies game c. battle of the sexes d. trust game

(c) Select all the Nash equilibria in pure strategies: a. DOWN–RIGHT b. DOWN–LEFT c. UP–LEFT d. UP–RIGHT

Exercise 3.3 – step by step

(a) Step 1. Rows: Up, Down. Columns: Left, Right. Each cell: (Row, Column).

Step 2. (Up, Left) $=$ $(8, 2)$; (Down, Right) $=$ $(4, 6)$; all other cells $(0, 0)$.

(b) Step 1. Both players earn something only if they coordinate (the diagonal), but Row prefers (Up, Left), 8 against 4, and Column prefers (Down, Right), 6 against 2.

Step 2. That is a **battle of the sexes** (c). Not a prisoner's dilemma: no player has a strategy that is best whatever the other does. Not matching pennies: there are pure-strategy equilibria. Not a trust game: moves are simultaneous.

(c) Step 1. Row: against Left, Up $8 >$ Down $0 \Rightarrow$ Up; against Right, Down $4 >$ Up $0 \Rightarrow$ Down.

Step 2. Column: against Up, Left $2 >$ Right $0 \Rightarrow$ Left; against Down, Right $6 >$ Left $0 \Rightarrow$ Right.

Step 3. Cells with both underlined: UP–LEFT and DOWN–RIGHT $\Rightarrow$ answers **a** and **c**.

Mixed strategies

A player randomises only if she is **indifferent** between the actions she mixes over.

p = Pr(Column plays Right), q = Pr(Row plays Up).

- Row’s expected payoffs depend on p; Column’s depend on q.
- Mixed equilibrium: p makes Row indifferent, $\mathbb{E}[\pi_{\text{up}}] = \mathbb{E}[\pi_{\text{down}}]$; q makes Column indifferent, $\mathbb{E}[\pi_{\text{left}}] = \mathbb{E}[\pi_{\text{right}}]$.

Best-response functions: Row plays Up ($q = 1$) on one side of the indifference value of p and Down ($q = 0$) on the other, and symmetrically for Column. Their intersections are *all* the Nash equilibria, pure and mixed.

Exercise 3.4

Expected profits and mixed strategies

Same game as Exercise 3.3. Let p denote the probability that the Column player chooses "right", and q the probability that the Row player chooses "up".

(a) Complete the following equations for the expected profits of each player's possible choices:

$$\mathbb{E}[\pi_{\text{up}}] = \square\,(1-\square) + \square \cdot \square \qquad \mathbb{E}[\pi_{\text{left}}] = \square \cdot \square + \square\,(1-\square)$$

$$\mathbb{E}[\pi_{\text{down}}] = \square\,(1-\square) + \square \cdot \square \qquad \mathbb{E}[\pi_{\text{right}}] = \square \cdot \square + \square\,(1-\square)$$

(b) The equilibrium in mixed strategies prescribes that the probability that Column plays right is $p = ?$, and the probability that Row plays up is $q = ?$

Exercise 3.4 – step by step

(a) Step 1. Row's payoffs depend on Column: Right with probability p, Left with probability $1-p$.

$$\mathbb{E}[\pi_{\text{up}}] = 8\,(1-p) + 0 \cdot p \qquad \mathbb{E}[\pi_{\text{down}}] = 0\,(1-p) + 4 \cdot p$$

Step 2. Column's payoffs depend on Row: Up with probability q, Down with $1-q$.

$$\mathbb{E}[\pi_{\text{left}}] = 2 \cdot q + 0\,(1-q) \qquad \mathbb{E}[\pi_{\text{right}}] = 0 \cdot q + 6\,(1-q)$$

(b) Step 3. Row mixes only if indifferent:

$$8\,(1-p) = 4p \;\Rightarrow\; 8 - 8p = 4p \;\Rightarrow\; 8 = 12p \;\Rightarrow\; p = \tfrac{2}{3}$$

Step 4. Column mixes only if indifferent:

$$2q = 6\,(1-q) \;\Rightarrow\; 2q = 6 - 6q \;\Rightarrow\; 8q = 6 \;\Rightarrow\; q = \tfrac{3}{4}$$

Step 5. Check: $p = \frac{2}{3}$: up $= 8/3$, down $= 8/3$. $q = \frac{3}{4}$: left $= 1.5$, right $= 1.5$.

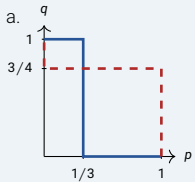
Typical mistake: solving Row's indifference for q. Row's payoffs only depend on p, so Row's indifference pins down p, Column's mix.

Exercise 3.5

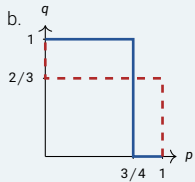
Best-response functions

Same game. Select the graph that gives the correct representation of the best-response functions (solid: Row's best response, dashed: Column's best response).

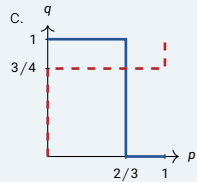
a.



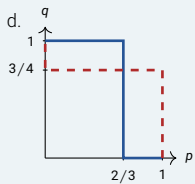
b.



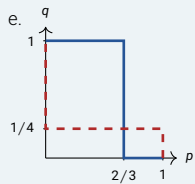
c.



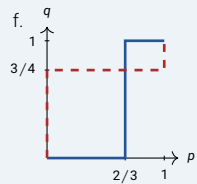
d.



e.



f.



Exercise 3.5 – step by step

Step 1. Row's best response as a function of p: Up if $8 - 8p > 4p \iff p < 2/3 \Rightarrow q = 1$ for $p < 2/3$, $q = 0$ for $p > 2/3$. The solid line steps *down* at $p = 2/3$.

Step 2. Column's best response as a function of q: Right if $6 - 6q > 2q \iff q < 3/4 \Rightarrow p = 1$ for $q < 3/4$, $p = 0$ for $q > 3/4$. The dashed line is at $p = 1$ below $q = 3/4$ and at $p = 0$ above it.

Step 3. Eliminate the wrong graphs:

- a: Row's threshold at $1/3$ b: thresholds swapped ($3/4$ and $2/3$)
- c: Column's line goes the wrong way e: Column's threshold at $1/4$
- f: both lines go the wrong way

Step 4. Answer **d**.

Step 5. Read the equilibria off the graph: the two lines meet at $(p, q) = (0, 1)$, Up–Left; at $(1, 0)$, Down–Right; and at $(2/3,\ 3/4)$, the mixed equilibrium.

PART 4

Multi-stage games

Strategies, Nash equilibrium, subgame perfection

Lecture 8 – “Strategies”, “Nash equilibria”, “Extensive form”, “Non-credible threats”, “Subgame-perfection”, “Backward induction”

- **Strategy**: a complete plan of action, covering every contingency, including decision points that are never reached.
- Lecture: with 2 proposals, the responder has 4 strategies (AA, AR, RA, RR), one reaction to each proposal.
- A **Nash equilibrium** may rest on a *non-credible threat*: (Equal, RA) is a NE, but rejecting the unequal split would reduce the responder’s payoff from $1 to $0.
- **Subgame perfect NE**: rationality in all parts of the game (subgames). Found by **backward induction**, starting from the final decision.
- Lecture: the only SPNE is (Unequal, AA).

Exercise 4.1

Bargaining – extensive form

Consider a game in which there is €6 to be divided, and the first mover is only permitted to make one of three proposals:

- LOW: €5 for the first mover and €1 for the second mover,
- EQUAL: €3 for each, or
- HIGH: €1 for the first mover and €5 for the second mover.

The second mover is shown the proposal and can either accept, in which case it is implemented, or reject and cause each to earn €0.

(a) Show this game in extensive form. How many subgames are there?

(b) How many strategies does each player have?

Exercise 4.1 – step by step

(a) **Step 1.** Root: the first mover, with three branches LOW, EQUAL, HIGH.

Step 2. At the end of each branch, a decision node of the second mover, with two branches A and R.

Step 3. Terminal nodes, payoffs (first, second): A after LOW $(\mathbf{5}, \mathbf{1})$, after EQUAL $(\mathbf{3}, \mathbf{3})$, after HIGH $(\mathbf{1}, \mathbf{5})$; every R gives $(\mathbf{0}, \mathbf{0})$.

Step 4. Subgames: a subgame starts at a decision node and contains everything that follows it. One for each of the second mover's three nodes, plus the whole game $\Rightarrow$ **4**.

(b) **Step 1.** First mover: one decision node, three actions $\Rightarrow$ **3** strategies.

Step 2. Second mover: a strategy must say what she does at *each* of her three nodes, including the ones that will not be reached: $2 \times 2 \times 2 = \mathbf{8}$.

Step 3. List them as (response to LOW, EQUAL, HIGH): AAA, AAR, ARA, ARR, RAA, RAR, RRA, RRR.

Typical mistake: answering "2 strategies" for the second mover (that is the number of *actions* at each node).

Exercise 4.2

Bargaining – a Nash equilibrium

Show that proposal HIGH in the game of Exercise 4.1 is part of a Nash equilibrium.

To finish specifying the decisions for this equilibrium, you have to say what the second mover's response is to each of the three proposals.

Exercise 4.2 – step by step

Step 1. For HIGH to be a best response of the first mover, LOW and EQUAL must pay her less than HIGH: the second mover must **reject** LOW and EQUAL, and **accept** HIGH. Candidate: (HIGH, RRA).

Step 2. First mover, given RRA: LOW $\rightarrow$ rejected, 0; EQUAL $\rightarrow$ rejected, 0; HIGH $\rightarrow$ accepted, 1. HIGH is a best response.

Step 3. Second mover, given HIGH: only the HIGH node is reached; accepting gives 5, rejecting 0, so she accepts. What she would do at the LOW and EQUAL nodes does not change her payoff, so RRA is a best response.

Step 4. Why not (HIGH, RRR)? Then the second mover could get 5 instead of 0 by accepting HIGH: not a best response.

Step 5. To find *all* Nash equilibria, write the 3×8 matrix and underline best responses:

	AAA	AAR	ARA	ARR	RAA	RAR	RRA	RRR
LOW	5, 1	5, 1	5, 1	5, 1	0, 0	0, 0	0, 0	0, 0
EQUAL	3, 3	3, 3	0, 0	0, 0	3, 3	3, 3	0, 0	0, 0
HIGH	1, 5	0, 0	1, 5	0, 0	1, 5	0, 0	1, 5	0, 0

First mover: max of each column. Second mover: max of each row. Seven cells with both underlined.

Exercise 4.3

Sequential rationality

(a) Does behaviour in the equilibrium discussed in Exercise 4.2 satisfy sequential rationality?
a. No b. Yes

(b) Find the subgame perfect Nash equilibrium of the game by backward induction.

Exercise 4.3 – step by step

(a) Step 1. Sequential rationality: each player must choose optimally at *every* decision node, including nodes that are not reached.

Step 2. Check RRA at the LOW node: reject gives 0, accept gives 1 $\Rightarrow$ rejecting is not optimal. Same at the EQUAL node: 0 against 3.

Step 3. The threat to reject LOW and EQUAL is not credible $\Rightarrow$ answer **a. No**.

(b) Step 1. Last stage first. LOW node: accept $1 > 0$. EQUAL node: accept $3 > 0$. HIGH node: accept $5 > 0$. $\Rightarrow$ second mover plays AAA.

Step 2. Replace each second-mover node with the payoff it leads to: LOW $\rightarrow (5, 1)$, EQUAL $\rightarrow (3, 3)$, HIGH $\rightarrow (1, 5)$.

Step 3. First mover chooses the largest first payoff: $5 > 3 > 1 \Rightarrow$ LOW.

Step 4. **SPNE: (LOW, AAA)**, payoffs $(5, 1)$. A strategy profile, not just an outcome: it says what the second mover does at every node.

Centipede game and noisy choice

Lecture 8 – “A treasure”, “A contradiction”, “The centipede game”, “The logit model”, “QRE and the non-credible threat”

Centipede (4 legs): backward induction $\Rightarrow$ Red stops at the first node, $(40, 10)$. McKelvey and Palfrey: only **8%** of Red players stop there.

Treasure vs contradiction: same SPNE (R, N), but when punishing (P) costs the second mover only 2, more than half of the first movers choose S.

Logit, with noise parameter λ:

$$\Pr(D_1) = \frac{\exp(\pi_1 \lambda)}{\exp(\pi_1 \lambda) + \exp(\pi_2 \lambda)}$$

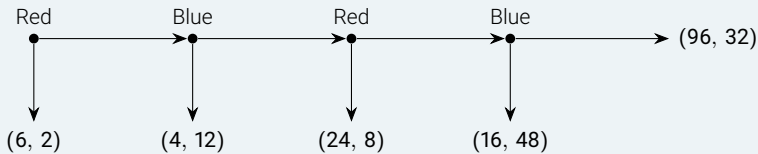
Lecture, contradiction game, $\lambda = 0.1$: $\Pr(P) = \dfrac{\exp(0.1 \times 48)}{\exp(0.1 \times 48) + \exp(0.1 \times 50)} = 0.45$.

QRE: better strategies are played more often, but not always; and expectations in equilibrium are correct.

Exercise 4.4

Centipede

At each node the player moving can *stop* (down), which ends the game, or *pass* (right). Payoffs are (Red, Blue).



(a) Find the subgame perfect equilibrium outcome by backward induction.

(b) What do experiments find in this type of game, and why?

Exercise 4.4 – step by step

(a) Start from the last node and move left.

Step 1. Node 4, Blue: stop $\rightarrow (16, 48)$, Blue gets 48; pass $\rightarrow (96, 32)$, Blue gets 32. $48 > 32 \Rightarrow$ **stop**.

Step 2. Node 3, Red: stop $\rightarrow (24, 8)$, Red gets 24; pass leads to Blue stopping, $(16, 48)$, Red gets 16. $24 > 16 \Rightarrow$ **stop**.

Step 3. Node 2, Blue: stop $\rightarrow (4, 12)$, Blue gets 12; pass leads to Red stopping, $(24, 8)$, Blue gets 8. $12 > 8 \Rightarrow$ **stop**.

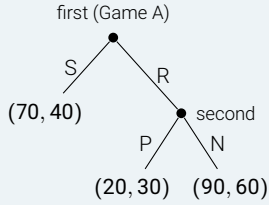
Step 4. Node 1, Red: stop $\rightarrow (6, 2)$, Red gets 6; pass leads to Blue stopping, $(4, 12)$, Red gets 4. $6 > 4 \Rightarrow$ **stop**.

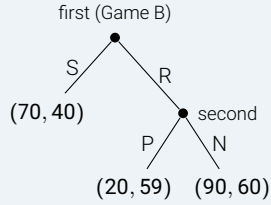
Step 5. Outcome: Red stops immediately, $(6, 2)$. Yet the total doubles at each pass: $8, 16, 32, 64, 128$.

(b) McKelvey and Palfrey: only 8% of Red players stop at the first node; most games end at node 2 or 3; 2% reach the end. Even a small amount of unpredictability in later decisions (for instance a few altruists) makes it optimal to continue early on.

Exercise 4.5

Noisy best responses





(a) Find the subgame perfect Nash equilibrium of each game. Is there a Nash equilibrium that is not subgame perfect?

(b) The second mover chooses according to the logit model with $\lambda = 0.1$. Compute the probability of P in each game.

Exercise 4.5 – step by step

(a) **Step 1.** Second mover: Game A, P 30 against N 60 $\Rightarrow$ N; Game B, P 59 against N 60 $\Rightarrow$ N.

Step 2. First mover, anticipating N: S gives 70, R gives 90 $\Rightarrow$ R. **SPNE (R, N)** in both games.

Step 3. (S, P) is a NE: if the second mover plays P, R gives the first mover $20 < 70$, so S is a best response; with S the second mover's node is not reached, so P is a best response too. It is not subgame perfect: P is a non-credible threat.

(b) **Step 1.** $\Pr(P) = \dfrac{\exp(\pi_P \lambda)}{\exp(\pi_P \lambda) + \exp(\pi_N \lambda)}$, with the second mover's payoffs and $\lambda = 0.1$.

Step 2. Game A: $\pi_P \lambda = 3$, $\pi_N \lambda = 6$; $\exp(3) = 20.1$, $\exp(6) = 403.4$:

$$\Pr(P) = \frac{20.1}{20.1 + 403.4} = 0.047.$$

Step 3. Game B: $\pi_P \lambda = 5.9$, $\pi_N \lambda = 6$; $\exp(5.9) = 365.0$, $\exp(6) = 403.4$:

$$\Pr(P) = \frac{365.0}{365.0 + 403.4} = 0.475.$$

Step 4. Same SPNE, but with noise P is almost never chosen in A and almost half of the time in B: why first movers choose S in the lecture's contradiction game.

PART 5

Oral questions

Sample oral questions I

Information cascades

1. Can you explain what an information cascade is?
2. Describe the experiment in which information cascades were studied.
3. Are information cascades compatible with the standard assumptions on perfect rationality, or do they only emerge if people are irrational?
4. Can a rational cascade induce the widespread adoption of a systematically incorrect choice?

Learning and simultaneous games

1. What is the difference between belief learning and reinforcement learning?
2. What happens to cooperation in a prisoner's dilemma repeated a known number of times, in theory and in the laboratory?
3. Describe the beauty contest game. What is its Nash equilibrium, and what do experiments show?

Sample oral questions II

Multi-stage games

1. What is a non-credible threat? Give an example.
2. What is the difference between a Nash equilibrium and a subgame perfect Nash equilibrium?
3. Describe the centipede game. What does backward induction predict, and what is observed?
4. How can a model of noisy best responses (logit, QRE) explain behaviour in games where the cost of punishment is very small?